# Brain alignment as a usable training signal: the internal extended manuscript

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**Abstract**

For a person and a language model reading the same text, a simple linear map predicts the person's brain activity from the model's middle-layer activations. For a decade this correspondence has been used in one direction only: to measure a model's alignment with the brain after it is trained. We ask whether it can instead be used during training. We test this in the setting of model compression: we distill a language model into a smaller student with brain alignment in the objective, at a matched budget, and ask whether the student preserves or improves that alignment, a question that remains untested. An information-budget argument implies that such a signal is far too small to act as a second teacher; at most it is a weak prior. Such a prior should help only where the text signal is weakest (small models, scarce data, aggressive compression) and wash out wherever data is abundant. **[GAP: Results summary deferred (checkpoint 2): the two established findings (the powered reality result and the no-preservation distillation screen) and the downstream arc (per-individual null, averaging artifact, quality law, predictability ceiling, no practical payoff) are summarised here only once R08–R14 are consolidated; close on the payoff line then.]**

## Contents

## Notation and terms

**Encoding model**
A ridge regression from a model's per-stimulus features to measured brain activity, scored out-of-sample. The standard brain-alignment instrument.

**Voxel / ROI**
A voxel is one 3D measurement location in an fMRI scan; an ROI is a predefined region (here, the five left-hemisphere language regions of the ROI screen).

**$\Delta R^2$ (unique $R^2$)**
The variance the language-model features add *on top of* the nuisance set: $R^2([Z_{\text{nuis}}, \text{LM}]) - R^2([Z_{\text{nuis}}])$. The semipartial $R^2$, not the partial. The only predictive score this work reports.

**Trained−untrained gap**
The verdict statistic: the unique $R^2$ of a language-trained model minus that of a same-architecture untrained network, under identical nuisance and splits. Isolates what *language training* adds over random initialisation.

**$Z_{\text{nuis}}$ (nuisance set)**
The low-level nuisance variables subtracted in both arms of the gap: utterance length, position, word and phoneme rate, word duration, log word-frequency, and a static word-embedding block (eng1000). Surprisal is deliberately excluded (it is itself LM-derived).

**Noise ceiling**
The maximum variance any model could explain, set by the reliability of the neural measurement itself; reported here as split-half reliability of a held-out repeated story.

**Permuted twin**
The per-kind shuffled-target null that tests brain-*specificity*. It belongs to the lever question (whether the signal can be *moved*), not to the reality question, and so does not appear in the Results section written here.

**$\mathbb{E}[Y \mid S]$**
The stimulus-predictable mean response: the strong conditioning variable behind the ceiling question, as opposed to the weak low-level proxy $Z_{\text{nuis}}$.

$\mathcal{L}_{\text{brain}}$
The brain-alignment training loss: the term added to a language objective that rewards representations predictive of recorded brain activity.

**Knowledge distillation (KD)**
Training a small student model to match a larger teacher's output distribution rather than the raw text labels. Pure logit KD is the perplexity-only form (temperature-scaled KL on the teacher's next-token logits, no hidden-state or brain term).

**Cold-init / warm-init KD**
A *cold-init* student is randomly initialised before KD, so any alignment it ends with was transmitted through the distillation channel; a *warm-init* student starts from a pretrained checkpoint and is therefore a fine-tuning-drift control, not a test of what KD carries.

**Headroom ($\Delta$)**
The teacher-minus-student gap in unique $R^2$, $\Delta = A_T - A_S$, with $A_T$ the teacher's alignment and $A_S$ the student's. The alignment a perplexity-only objective leaves below the teacher.

**Floor-anchored retention ($\rho'$)**
The fraction of the teacher's alignment a student keeps above the untrained floor, $\rho' = (A_S - A_0)/(A_T - A_0)$, where $A_0$ is the untrained score: $\rho' = 0$ is the floor (destroyed), $\rho' = 1$ is the teacher (preserved). Distinct from the partial correlation $\rho_{XY \cdot Z}$ of the methods (the primed symbol is retention, never a correlation).

**Data-processing inequality (DPI)**
For a Markov chain $A \to B \to C$, no processing of $B$ can increase information about $A$: $I(A;C) \leq I(A;B)$. Here it bounds a fixed-function compression to lose, never create, alignment relative to its teacher.

# 1 Introduction

For a decade, one instrument has anchored the study of language and the brain. A person and a language model read the same text. A linear map predicts the person's recorded brain activity from the model's middle-layer activations. Its accuracy on held-out text is the model's brain-alignment score [1].

This map almost always runs in one direction. It *measures* alignment, as a property read off a model after training. We ask whether it can run the other way. Can alignment instead be *used* during training, as a term in the objective that rewards representations from which brain activity is linearly predictable? And if it can, does a model trained that way gain anything a text-only model does not?

**Problem.** The concrete setting is model compression. Knowledge distillation trains a small *student* model to imitate a larger *teacher*, refitting the student's representations from scratch. That refit tends to discard the structured middle-layer representations where brain alignment lives. The problem this thesis studies is whether brain alignment, used as a training signal during distillation, can preserve or even improve it in the student at a matched compute budget, and whether the result buys anything beyond the alignment score itself. The case matters because compression is how small, deployable models are built, and the recent literature leaves it untested.

**Hypothesis.** If a brain signal helps at all, it helps as a weak prior, not as a second teacher. The reason is a mismatch in scale. A language model learns from on the order of a trillion tokens of text. A language-fMRI dataset offers a few thousand noisy recordings, and a low noise ceiling caps the variance any model can explain in them. A prior that small cannot teach task structure the text does not already contain. What it can do is select among the many

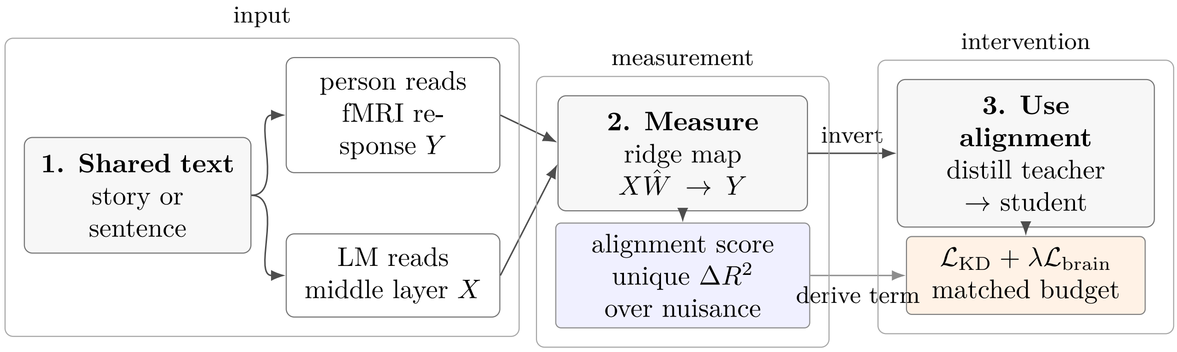


Figure 1: The measurement instrument and its inversion. A ridge encoding model maps language-model activations to brain responses; held-out accuracy is reported as unique variance over the nuisance floor. The thesis asks whether a corresponding brain-alignment term can guide a distilled student.

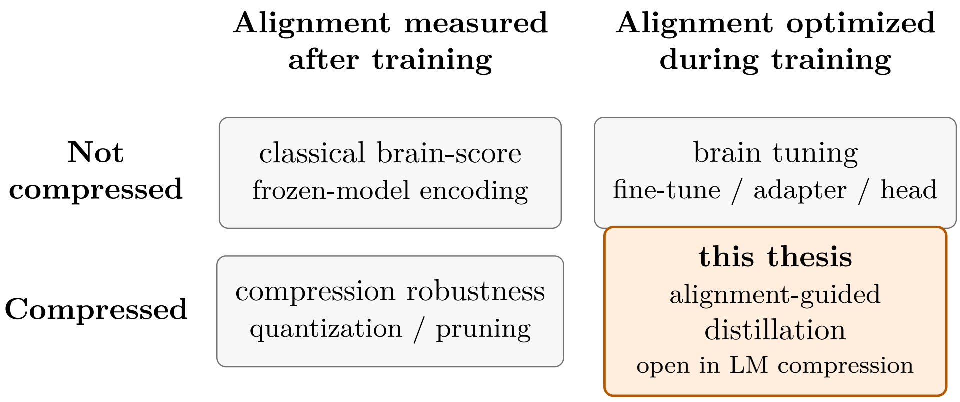


Figure 2: The field on two axes; three cells are occupied and the fourth, where this work sits, is open in language-model compression. Examples and citations are given in Table 1.

text-consistent solutions a large model admits, the way a weak regularizer does. This gives a falsifiable prediction. A brain prior should help most where the text signal is weakest, in small, heavily compressed, or data-starved models, and should wash out wherever text data is abundant. Section 3 makes the argument precise.

Using fMRI to shape a language model is not new. It has been done for speech models [2] and for text models [3], and its effect is causal: text models pushed to be brain-*misaligned* at matched perplexity perform worse across a broad downstream suite [4]. Every one of these methods *grows* the model, by fine-tuning it, adding adapters, or attaching a brain-prediction head, and none compresses it. The single study that compresses language models only measures how quantization and pruning move an alignment score. It never puts alignment in the objective, and never tests distillation [5]. The work splits along two questions: does an intervention grow or shrink the model, and does it measure or optimize alignment? Three of the four combinations are occupied, and one is empty (Figure 2). This thesis occupies the empty cell. It studies alignment-guided distillation at a matched student budget, the regime where our hypothesis says a brain prior could matter, rather than one where it is assumed to pay off.

Brain-alignment scores are easy to inflate. Shuffled splits leak through the temporal autocorrelation of fMRI, and low-level features such as sentence length carry much of an apparent score, so an untrained network can appear to explain brain data on its own [6]. We therefore hold every number in this thesis to a fixed standard, set before measuring anything. The contributions below summarize that standard, and Section 3 defines it [1, 7].

**Contributions.** This thesis makes the following contributions, scoped to the results it establishes here.

- It formulates alignment-guided distillation and locates it as the one combination the recent literature leaves open (Figure 2).
- It gives an information-budget argument that predicts where a brain prior can and cannot help, turning a vague argument for using brain data into a falsifiable prediction.
- It fixes a measurement protocol that controls for the artifacts that can fake an alignment score: unique variance over a capacity-matched nuisance floor, contiguous splits, a same-architecture untrained control, a held-out noise ceiling, and a permuted-target control for specificity, the controls the alignment-training literature has largely lacked.
- It establishes, at voxel scale and with statistical power, that the alignment signal is real beyond those artifacts [E006].
- It shows that plain perplexity-only distillation does not preserve alignment by default, while making explicit that the drop cannot yet be separated from the language-model quality the student loses [E003].
- It shows that the apparent alignment-objective effect is small, entangled with model quality, and does not survive the honest unit of inference [E004].

The program asks five questions in sequence, each only if the previous one holds. Is the signal real, or an artifact? Does plain distillation preserve it? Does an alignment objective move it beyond what perplexity already implies? Does any gain survive per individual at matched perplexity? And does it improve a downstream task?

The first three are settled here: the signal is real [E006], plain distillation does not preserve it [E003], and an alignment objective moves it only weakly and in step with model quality [E004].

# 2 Related work

This thesis sits in the one empty cell of a two-by-two grid: grow or shrink the model, and measure or optimize alignment (Table 1). We review the three occupied cells in turn, then the measurement standard this work adopts.

Table 1: The field on two axes: what an intervention does to the parameter budget, and how it treats brain alignment. Three cells are occupied; the fourth, shrinking a model while alignment is in the objective, is empty in the 2026 literature and is where this work sits.

| | Alignment measured | Alignment optimized |
|---|---|---|
| Model grown | classical brain-score [1, 8, 9] | brain-tuning [2–4] |
| Model shrunk | compression robustness [5] | **this thesis (empty in the literature)** |

## 2.1 The classical brain-score literature

The largest body of work fits encoding models to frozen, pretrained networks, then reads off how alignment varies with architecture, scale, and training. Three findings recur. Alignment lives in the structured middle layers, where syntax drives the layerwise trend and semantics is more region-specific [1]. It grows with model scale and saturates around a few billion parameters, and at a matched parameter count scale moves it more than instruction-tuning does, which is why a base model is the right teacher to align against [8]. Beyond that scale the trend can flatten or reverse, as the largest models outgrow the human language network and track formal competence more than behavior [9]. This line of work shows the signal exists and is structured, but it never intervenes through alignment, so it cannot say whether alignment is usable.

## 2.2 Brain-tuning

A second line puts alignment in the objective of a fine-tuned or grown model, across modalities. For speech, a voxelwise brain loss fine-tunes compact encoders and improves both alignment and several downstream tasks, with a multi-participant variant reporting further gains at higher data efficiency [2, 10]. For text, training GPT-2 and LLaMA-2 with a cosine brain loss beats text-only baselines, though the gains are mixed and the splits are shuffled with no brain-shuffled control, so they would not clear the measurement bar this work adopts [3]. The strongest evidence is causal. Training a text model to be *less* brain-aligned, at fixed perplexity, makes it worse across a suite of more than two hundred tasks, while training it to be *more* aligned improves performance on the same suite [4]. Alignment is therefore doing causal work, not just riding along with a good model. The bare inverted idea is not novel, but no study here compresses a model: every method grows or holds the parameter budget, leaving the compression question open.

## 2.3 Alignment under compression

The closest prior work to this thesis is the compression study of Oota et al. [5], which tracks how alignment changes as a model is compressed. They find that alignment survives only certain post-training quantizers on models of at least a few billion parameters; other quantizers, smaller models, and pruning all degrade it, and distillation is never tested. That study measures alignment under compression but never optimizes it. It reports how quantization and pruning move the score, and never makes alignment a training target. Distillation is a different and lossier operation. It refits a smaller model from scratch against the teacher, the step most likely to destroy the middle-layer structure where alignment lives. So alignment surviving quantization, a gentle reshaping of fixed weights, does not imply it survives distillation. Whether distillation preserves alignment is therefore an open empirical question: how much does it lose by default, and can a brain objective recover the rest? We develop the information-theoretic bound behind this in Section 3.

## 2.4 The empty cell

No prior work occupies the empty cell. Two nearby precedents, one in vision and one in audio, are sometimes cited as indirect support for a brain regularizer when data is scarce, but neither isolates a brain-specific effect. Pirlot et al. [11] regularize a vision network with neural data and improve both accuracy and adversarial robustness, yet a shuffled-label control reproduces the accuracy gain, so only the robustness benefit depends on real neural structure. Freteault et al. [12] fine-tune a small audio network on fMRI and improve most downstream tasks, but without an interaction test or a matched non-brain control, the low-data advantage is not isolated. Neither study is in the distillation regime this work targets, so both inform our design without testing the claim.

## 2.5 The measurement bar

A separate line of work warns that alignment scores are easy to inflate and over-read, and it sets the standard this work adopts. Shuffled splits leak through the temporal autocorrelation of fMRI, low-level features like sentence length carry much of the apparent signal, and some published scores fail basic predictive and reliability checks [6, 13]. The same headline score can reflect real semantics in one model and shallow phonology in another, so the number alone does not pin down the mechanism [14]. A differentiable, fMRI-free surrogate would go further, but the literature does not agree on its direction: local intrinsic dimension correlates with alignment in opposite ways across language and vision, and similarity-based surrogates track width and depth as much as alignment [15–17]. We therefore leave a surrogate to future work, since real fMRI must first establish which way to push it.

# 3 Methods

## 3.1 The alignment score

We measure alignment with a ridge encoding model that maps a frozen model's per-stimulus middle-layer activations $X$ to recorded brain activity $Y$, fit on training stimuli and scored on held-out ones. For activations $X \in \mathbb{R}^{n \times d}$ over $n$ stimuli and a $v$-voxel response $Y$, the fitted map is

$$\hat{W} = \arg\min_{W} \|Y - XW\|_F^2 + \lambda \|W\|_F^2, \tag{1}$$

and the score for each voxel is the correlation between its predicted and measured response. A raw score of this kind means little on its own, because a model can predict the brain for reasons that have nothing to do with language: utterance length, word position, generic word identity. We therefore report not the raw score but the variance the model adds *on top of* a low-level nuisance set $Z_{\text{nuis}}$, the difference of two cross-validated fits,

$$\Delta R^2 \;=\; R^2\big([Z_{\text{nuis}}, \text{LM}]\big) - R^2\big([Z_{\text{nuis}}]\big). \tag{2}$$

A positive $\Delta R^2$ says the contextual representation explains brain activity that length, rate, and static word identity cannot [E002].

## 3.2 What the score measures

We use the score as a linear lower-bound proxy for an information increment: the information the model's representation carries about brain activity beyond the low-level baseline. The underlying estimand is conditional mutual information. For three variables $(X, Y, Z)$,

$$I(X; Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z) \tag{3}$$

is the dependence between $X$ and $Y$ that remains once $Z$ is known [18]. Conditioning is essential here. Plain mutual information cannot distinguish the dependence of interest from a shared driver: if $X$ and $Y$ are both driven by some common $Z$, then $X \perp Y \mid Z$, so the conditional term is zero even when the unconditional term is positive. The chain rule splits the dependence into a baseline and the increment over it,

$$I\big(B; [Z_{\text{nuis}}, \text{LM}]\big) = I(B; Z_{\text{nuis}}) + I(B; \text{LM} \mid Z_{\text{nuis}}), \tag{4}$$

with $B$ the brain response, so the information the model's representation adds over the nuisance is exactly the conditional term $I(B; \text{LM} \mid Z_{\text{nuis}})$.

The conditional mutual information is not directly measurable from data, so we estimate it through the unique-variance score, which is its linear-Gaussian surrogate. Under joint Gaussianity, conditional mutual information is a monotone function of the squared partial correlation,

$$I(X; Y \mid Z) = -\tfrac{1}{2} \ln\big(1 - \rho^2_{XY\cdot Z}\big), \tag{5}$$

so the quantity is positive exactly when the partial correlation is nonzero [18, Ch. 8]. The reported score is the *semipartial* $R^2$, the raw increment, which differs from the partial $\rho^2$ only by the factor $1/(1 - R^2([Z_{\text{nuis}}]))$. Both vanish together, so a positive semipartial score is equivalent to a positive conditional mutual information. Ridge is linear, so the score captures the linearly decodable part of that information, a population-level lower bound.

The choice of $Z$ fixes what the claim means. Conditioning on the low-level nuisance $Z_{\text{nuis}}$ shows the signal is real beyond low-level features, not beyond the stimulus as a whole. Conditioning instead on the full stimulus-predictable response $\mathbb{E}[Y \mid S]$ asks a stronger question: not whether the representation explains brain activity beyond low-level features, but whether it explains anything the stimulus itself does not already predict. That is the stimulus-predictability ceiling, and it reuses the same chain rule with $\mathbb{E}[Y \mid S]$ in place of $Z_{\text{nuis}}$ as the larger conditioning set.

### 3.3 Why a brain signal is only a weak prior

A brain term in a language objective is, up to constants, a maximum-a-posteriori estimate. Training chooses parameters $\theta$ that minimise a loss, and adding a brain term to the usual language-model loss gives

$$\hat{\theta} \;=\; \arg\min_{\theta} \; \underbrace{\Big[ -\textstyle\sum_{t=1}^{N} \log p_{\theta}(x_t \mid x_{<t}) \Big]}_{-\log p(\mathcal{D}_{\text{text}} \mid \theta)} \;+\; \lambda \; \underbrace{\mathcal{L}_{\text{brain}}(\theta)}_{-\log p(\theta)}. \tag{6}$$

The first term is the negative log-likelihood of the text: next-token prediction is maximum likelihood, and minimising it maximises $p(\mathcal{D}_{\text{text}} \mid \theta)$. The second term is the negative log-prior: any regulariser that penalises a property of $\theta$ corresponds, up to a normalisation constant, to a prior $p(\theta) \propto \exp\big(-\lambda \mathcal{L}_{\text{brain}}(\theta)\big)$ over parameters that favour that property, with $\lambda$ setting its strength. Minimising their sum is therefore

$$\hat{\theta}_{\text{MAP}} \;=\; \arg\max_{\theta} \; \big[ \log p(\mathcal{D}_{\text{text}} \mid \theta) + \log p(\theta) \big], \tag{7}$$

a MAP estimate in which the text is the likelihood and the brain term is the log-prior. But how far can a prior that small move a posterior dominated by the text?

A training corpus carries on the order of $10^{12}$ to $10^{13}$ bits of supervision. The brain side offers a few thousand noisy stimuli, and the information each voxel can carry is capped by its noise ceiling, a channel capacity

$$I \leq \tfrac{1}{2} \log_2(1 + \text{SNR}), \qquad \text{SNR} = \frac{r_{\max}^2}{1 - r_{\max}^2}, \tag{8}$$

which puts the usable brain budget, as an order of magnitude, around $10^4$ to $10^6$ bits [18]. The data-processing inequality shrinks it further. Let $S$ be the stimulus and $B$ its noisy brain response. Under the modelling assumption that $B$ is generated from $S$, the chain $\theta^\star \to S \to B$ gives

$$I(\theta^\star; B) \leq I(\theta^\star; S). \tag{9}$$

The brain response cannot contain more information about the target parameters than the stimulus that elicited it. Such a small prior cannot teach new task structure the text does not already contain. It can only select among the many near-equivalent solutions an overparameterized model admits, which defines the role of the brain term as a regularizer rather than a source of new information.

A generalization bound can show quantitatively how and where such a weak prior can help. Modelling training as a channel from the data $Z^n$ to the learned parameters $\theta$, the expected generalization gap is bounded by the mutual information between them,

$$\overline{\text{gen}} \;\leq\; \sqrt{\frac{2\sigma^2\, I(\theta; Z^n)}{n}}, \tag{10}$$

for $\sigma$-sub-Gaussian loss [19]. If the brain prior adds $\Delta I$ bits to $I(\theta; Z^n)$, its benefit to generalization is at most $O(\sqrt{\Delta I / n})$, and it shrinks as the task data $n$ grows. A weak but well-targeted brain prior should therefore help most where the text signal is weakest, in small, under-trained, or aggressively compressed models, and fade where text data already dominates. Distillation is the regime where that prediction is most testable.

**[DIAGRAM PLACEHOLDER: the unique-variance score and the trained–untrained gap]**

*A small construction diagram. Top: for one network, two nested cross-validated ridge fits, $R^2([Z_{\text{nuis}}])$ and $R^2([Z_{\text{nuis}}, \text{LM}])$; their difference is the unique-variance score $\Delta R^2$ (the increment the representation adds over the nuisance floor). Bottom: the same construction run twice, once for the trained network and once for the same-architecture untrained one; the reported statistic is the* gap, *trained $\Delta R^2$ minus untrained $\Delta R^2$, so the shared nuisance term cancels. Render in TikZ as two stacked bar-pairs with the increment and the gap annotated.*

Figure 3: The two reported quantities: the unique variance a network adds over the nuisance floor, and the trained-minus-untrained gap.

### 3.4 The trained–untrained gap

We report the *gap* between a trained model and a same-architecture untrained network, both fit with the same nuisance set and the same splits, rather than the trained model's absolute score. An absolute score is insufficient on naturalistic data, because even an untrained network can score slightly positive. The stimulus drives both the brain response and the untrained features; if the nuisance set captures only part of the stimulus, residual shared-cause dependence can remain. Subtracting the untrained arm removes the component shared by architecture, tokenisation, and residual temporal structure. The gap therefore estimates what *language training* adds over random initialisation [E006].

### 3.5 The controls

We apply three controls uniformly. *(1) Contiguous splits*: train and test are divided in blocks (block cross-validation on the ROI screen, whole-story-held-out cross-validation on the voxelwise run), because shuffling time-adjacent fMRI samples leaks near-duplicate timepoints across the split and inflates the score [6][E002][E006]. *(2) A capacity-fair nuisance floor*: the wide contextual block and the wide static-embedding block are each reduced by PCA to the same rank, fit on the training fold only, so the language model does not win merely by carrying more dimensions; the few low-level scalars stay raw [6, 13][E006]. *(3) A held-out noise ceiling*: on the voxelwise run only reliable voxels are kept, selected by split-half reliability on a separate story repeated ten times and held entirely out of the cross-validation pool, so the estimate is not inflated by voxels that happen to correlate with the model by chance [20, 21][E006]. Surprisal is deliberately left out of the nuisance set, since it is itself LM-derived and subtracting it would remove part of the construct under test; log word-frequency, a static lexical scalar, is distinct and is subtracted [E006]. The permuted-target null shuffles brain responses across stimuli, breaking the stimulus-response pairing while preserving temporal structure [2, 11]. It tests brain-*specificity*, not whether the baseline alignment signal is real, and enters only where the question is whether the signal can be moved.

### 3.6 Data and models

We use two datasets. The ROI screen uses the 1000 isolated sentences and five left-hemisphere language regions of Tuckute et al. [22]. The ROI benchmark averages responses across training participants, so it is coarse, but its isolated-sentence design avoids the temporal-autocorrelation leakage of naturalistic time series; its reported noise ceiling is $r \approx 0.35$ [E002]. The powered measurement uses subject UTS03 of the deep-sampled naturalistic story-listening dataset of LeBel et al. [21]: 20 stories, story-level folds, and 11,442 voxels retained at split-half reliability above 0.5 on a held-out repeated story [E006]. The models are GPT-2, GPT-2-medium, and Qwen2.5-0.5B, each read at its alignment-peak middle layer, with a same-architecture untrained network as the control in every comparison [E002][E006].

### 3.7 Distillation and the brain term

The compression setting is knowledge distillation: a smaller *student* learns to reproduce a larger *teacher*'s next-token distribution, rather than learning from text directly. Perplexity gives the output-quality control. Per-token cross-entropy is code length, maximum-likelihood training minimizes the KL divergence from the data distribution to the model's, and perplexity is the exponential of that cross-entropy [18]. Both objectives are KL objectives with different references: plain language-model training projects the student onto the empirical data, while distillation projects it onto the teacher's softened distribution. Matching *held-out perplexity* across arms holds output quality fixed while the objective varies; matching compute does not.

The distillation objective does not by itself imply preserved alignment, for two reasons. First, it constrains the *output* distribution, whereas alignment is a property of the *internal* representation, and the readout from representation to logits is many-to-one,

$$D\big(p_T \,\|\, q_S\big) \to 0 \;\not\Rightarrow\; Z_S \approx Y_{\text{teacher}} \;\not\Rightarrow\; I(Z_S; B \mid Z_{\text{nuis}}) = I(Y_{\text{teacher}}; B \mid Z_{\text{nuis}}), \tag{11}$$

so matching the teacher's output need not match its representation. Second, a from-scratch student also fits the stimulus through the raw next-token targets, a route the teacher's features do not mediate. That route breaks the Markov chain behind the data-processing bound, so the teacher's alignment no longer upper-bounds the student's. Whether distillation preserves alignment is therefore an empirical question, not a corollary.

The brain term enters the objective as a co-trained regression loss,

$$\mathcal{L} = \mathcal{L}_{\text{KD}} + \lambda\, \mathcal{L}_{\text{brain}}, \tag{12}$$

with mixing weight $\lambda$. The verdict-bearing arm is a randomly initialised student, because any alignment it retains must be learned during distillation rather than inherited from pretraining. A warm-initialised student and a no-teacher fine-tune are kept as labelled drift controls, and every arm is scored at one fixed layer within a single architecture so no layer choice biases the comparison [E003]. Brain *specificity* is tested with a permuted-target twin: the same loss, but against brain targets shuffled across stimuli. The arms are matched by compute budget, not by optimizer-step count; the from-scratch arm runs about twice the optimizer steps of the warm arms, a caveat carried into the results [E003].

## 4 Results

### 4.1 The alignment signal is real beyond low-level artifacts

Under contiguous splits and the full low-level nuisance regression, a trained model's middle layer carries unique, out-of-sample variance about language-network activity that a same-architecture untrained network does not. The ROI screen shows the effect across models, and the powered voxelwise run confirms the effect at voxel scale.

The ROI screen uses the five left-hemisphere language regions of Tuckute et al. [22], 1000 isolated sentences, and five contiguous folds. The screen shows a positive trained-minus-untrained gap for all three models, with the untrained control negative at every layer and seed (Table 2). The signal is positive across a broad plateau of middle layers rather than at a single peak, and is absent from random features [E002].

Table 2: ROI screen on Tuckute 2024 (five LH language ROIs, five contiguous folds). Unique $R^2$ is the semipartial increment over the nuisance set; the gap is trained minus untrained. NC-norm is the gap as a fraction of the $r \approx 0.353$ noise ceiling.

| Model | Best layer | Trained $\Delta R^2$ (mean $\pm$ SD) | Untrained $\Delta R^2$ | Gap | NC-norm | Source |
|---|---|---|---|---|---|---|
| GPT-2 (12L) | L7 | $+0.020 \pm 0.008$ | $-0.011$ | $+0.030$ | 0.056 | [E002] |
| GPT-2-medium (24L) | L14 | $+0.019 \pm 0.005$ | $-0.014$ | $+0.033$ | 0.053 | [E002] |
| Qwen2.5-0.5B (24L) | L12 | $+0.036 \pm 0.010$ | $-0.014$ | $+0.050$ | 0.102 | [E002] |

The powered voxelwise run uses subject UTS03 of LeBel et al. [21], 20 naturalistic stories, and 11,442 voxels selected at split-half reliability above 0.5 on a held-out repeated story. The voxelwise run gives a trained-minus-untrained gap of +0.0207 [95% CI +0.0205, +0.0209] for GPT-2 and +0.0277 [95% CI +0.0274, +0.0280] for Qwen2.5-0.5B; 95% and 99% of reliable voxels are positive respectively, and the untrained unique $R^2$ is $-0.017$ in both (Table 3) [E006]. The confidence intervals exclude zero, and the gap is carried by language training rather than by architecture or the temporal structure the nuisance floor absorbs [E006].

Table 3: Powered voxelwise run on LeBel UTS03 (20 stories, story-level folds). The gap is the trained-minus-untrained unique $R^2$; CIs are bootstrap.

| Model | Layer | Trained − untrained gap [95% CI] | % reliable voxels gap > 0 | Source |
|---|---|---|---|---|
| GPT-2 | L7 | +0.0207 [95% CI +0.0205, +0.0209] | 95% | [E006] |
| Qwen2.5-0.5B | L12 | +0.0277 [95% CI +0.0274, +0.0280] | 99% | [E006] |

Three controls address the main artifacts, with complementary roles in the screen and the powered run. The ROI screen uses isolated sentences. The isolated-sentence design removes temporal autocorrelation from naturalistic time series, so a positive result is harder to obtain rather than easier [E002]. The voxelwise run uses naturalistic stimuli but holds out whole stories, so test timepoints are never adjacent to training timepoints. Voxel selection also uses a separate repeated story, preventing a score driven by voxels that correlate with the model by chance [E006]. Both the trained and untrained scores subtract the low-level set most likely to produce a false positive: word and phoneme rate, word duration and length, log word frequency, and a static word-embedding block [E006]. Surprisal and imageability are deliberately left unsubtracted, so the floor alone does not rule out residual contextual structure. The untrained network provides the stronger check: under the same architecture and the same stimulus set, the control scores negative at $-0.017$, so architecture, rate, and residual temporal structure cannot explain the gap [E006].

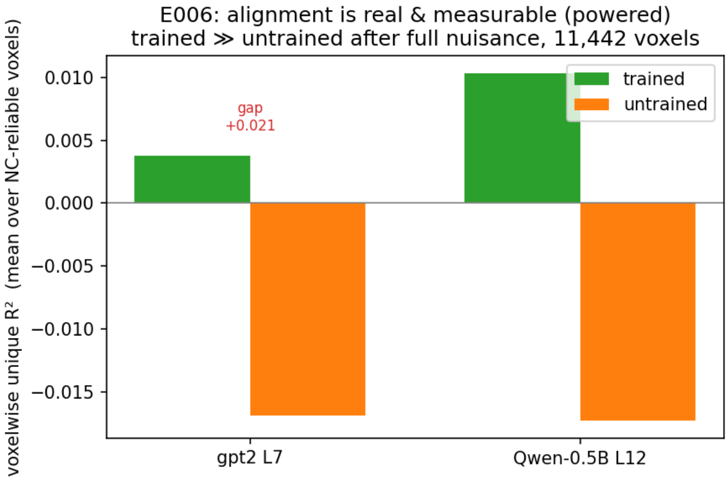


Figure 4: The powered reality result: the trained-minus-untrained unique-$R^2$ gap is positive across the overwhelming majority of reliable voxels on LeBel UTS03 [E006].

The alignment signal is real beyond low-level artifacts in neural data. The three-model ROI screen supplies breadth, and the deeply sampled single subject UTS03 supplies voxel-scale statistical power. The single-subject design leaves cross-subject generality of the powered effect open. Qwen aligns more strongly than either GPT-2 variant, while the two GPT-2 variants are nearly tied despite a threefold parameter difference. The single cross-family contrast does not by itself establish a quality ordering; the quality-ordering question is taken up later [E002]. The Q0 result is necessary for the rest of the program, but not sufficient for it. The Q0 result establishes that the signal is real, not that the signal can be moved or that moving it improves a model.

## 4.2 Plain distillation does not preserve alignment by default

Knowledge distillation (KD) trains a smaller student to match a larger teacher's output distribution rather than to learn directly from text. A distillation-trained student does not keep the teacher's brain alignment by default. When arms are ordered by how much of each representation was rebuilt through the distillation channel rather than inherited from ordinary pretraining, unique $R^2$ falls monotonically. A from-scratch logit-KD student lands far below its teacher, a gap of $+0.0181$ ($p < 0.001$) at which it retains only 0.37 [0.14, 0.58] of the teacher's alignment above an untrained floor [E003]. The cold-init drop rules out automatic preservation, the outcome under which compression would need no alignment intervention, and measures the headroom a perplexity-only objective leaves below the teacher. The E003 result does not establish the stronger claim that the distillation objective reduces alignment beyond the language-model quality it costs. Across the trained and distilled models, alignment tracks log-perplexity at Pearson $-0.88$, so quality alone could account for the whole gradient [E003][L011].

The headroom is $\Delta = A_T - A_S$, where $A_T$ is the teacher's alignment and $A_S$ is the student's alignment. Both quantities are measured with the same protocol as the reality result: unique $R^2$ on the Tuckute ROI screen, contiguous folds, and the capacity-fair nuisance floor. Because preservation is empirical rather than guaranteed, the design needs an arm whose alignment can only arrive through distillation. For deterministic post-processing of a fixed teacher, such as the quantization and pruning of Oota et al. [5], the student's features depend on the stimulus only through the teacher's. The data-processing inequality then bounds $I(Z_{\text{student}}; B) \leq I(Y_{\text{teacher}}; B)$, so a fixed-function compression cannot exceed the teacher's alignment under the teacher-to-student information path. In the fixed-function setting, the study asks how much alignment is lost. Distillation lacks a fixed-function path. A from-scratch stu-

Missing figure: ../figures/fig_kd_retention.png

Figure 5: [PLACEHOLDER: generate from [E003]] Two panels. (A) Floor-anchored retention $\rho'$ along the rebuild ordering (teacher, conventional GPT-2, warm-init KD, distilgpt2, cold-init KD, untrained floor) with bootstrap CIs, showing the monotone fall. (B) Unique $R^2$ versus log-perplexity across the same arms, with the Pearson $-0.88$ fit, showing that alignment co-varies with model quality. **[GAP: Round-3: render the two-panel figure (scripts/figures) and finalise the caption.]**

dent fit by gradient descent on the teacher's logits has an additional route through the training text, bypassing the teacher's features. The data-processing premise fails, and nothing forces the student back to the teacher's alignment [E003].

The verdict rests on a cold-init student. A warm-started student inherits full teacher-level alignment from its own pretraining, which the reality result already measures. High retention for the warm-started arm would show only that a few KD epochs did not erase existing alignment, not that the KD channel carried the alignment [E003]. The verdict-bearing arm is a randomly initialised GPT-2 trained on the teacher's logits alone, the only arm whose surviving alignment must pass through distillation. The warm-init KD arm and the no-teacher plain-LM finetune remain as labelled fine-tuning-drift and corpus-drift controls. Every arm is scored at one fixed layer within the 12-layer GPT-2 architecture, so cross-architecture layer choice cannot bias the comparison [E003].

Table 4: Knowledge-distillation arms on the Tuckute ROI screen at the fixed verdict layer (NC-normalised unique $R^2$, noise ceiling 0.353) [E003]. Floor-anchored retention $\rho' = (A_S - A_0)/(A_T - A_0)$ is 0 at the untrained floor and 1 at the teacher; $\Delta$ is the gap versus the teacher; $p$(no-drop) is the bootstrap probability of no drop from the teacher.

| Arm | Init / objective | $\Delta R^2$ | $\rho'$ [boot. CI] | $\Delta$ vs teacher | $p$(no-drop) | ppl | Source |
|---|---|---|---|---|---|---|---|
| teacher gpt2-medium | pretrained | +0.0188 | n/a | n/a | n/a | 74 | [E003] |
| gpt2 | pretrained (conventional) | +0.0195 | 1.02 [0.77, 1.27] | $\approx 0$ | 0.57 | 105 | [E003] |
| lmft_warm | warm + plain-LM finetune | +0.0174 | 0.95 [0.81, 1.13] | +0.0016 | 0.265 | 44 | [E003] |
| kd_warm | warm + pure logit KD | +0.0144 | 0.84 [0.67, 1.04] | +0.0046 | 0.061 | 95 | [E003] |
| distilgpt2 | real distillation (+hidden-cosine) | +0.0076 | 0.60 [0.42, 0.84] | +0.0119 | 0.001 | 169 | [E003] |
| **kd_cold** | **random + pure logit KD** | +0.0005 | 0.37 [0.14, 0.58] | +0.0181 | 0.000 | 477 | [E003] |
| untrained | random (floor) | −0.0102 | 0 | n/a | n/a | ~50000 | [E003] |

The supported finding is the monotone gradient (Table 4). Retention falls monotonically along the rebuild ordering: conventional GPT-2 at the teacher's level ($\rho' \approx 1.0$), warm-init KD at 0.84, the published distilgpt2 at 0.60, the from-scratch cold-init student at 0.37, and the untrained floor at 0 [E003]. The reference arms validate the measurement: off-the-shelf GPT-2 sits at the teacher's level and the untrained control at the floor, and two runs scoring the same references on different GPUs reproduced the reference scores to about 0.0003 [E003]. The ordering holds at every PCA rank tested ($n_{\text{pca}} \in \{25, 50, 100\}$), but the absolute loss fractions are rank-sensitive and the two quality-entangled arms change sign in unique $R^2$ across ranks, so the ordering is the finding and the magnitudes are rank-dependent [E003].

The gradient does not isolate the distillation objective from the language-model quality the student loses, because alignment co-varies with quality across the same arms. Alignment is tightly predicted by log-perplexity across these arms (Pearson −0.88). In the log-perplexity fit, warm-init KD lies essentially on the regression line, while conventional GPT-2 is the only arm more aligned than its perplexity predicts. The two contrasts that would separate an objective-specific loss from "alignment tracks quality" are both marginal. Warm-init KD is less aligned than GPT-2 despite lower perplexity, but the bootstrap $P(\text{gpt2} \le \text{kd\_warm}) = 0.092$. At matched budget, the no-teacher finetune beats warm-init KD on both perplexity and alignment, but $P(\text{lmft} \le \text{kd\_warm}) = 0.085$. Each contrast leans on a single unreplicated point and, under a paired-fold test, on one fold carrying about 44% of the signal [E003][L011]. Under-training adds another caveat: the cold-init arm's held-out perplexity (477) sits far above the converged arms' (44 to 95). Part of its near-floor alignment may reflect a worse language model rather than distillation reducing alignment. The design predeclared this convergence caveat [E003]. The arms were matched on corpus, batch size, learning rate, and architecture, but not on optimizer-step count: the from-scratch arm ran about twice the steps of the warm arms, and the step imbalance deepens rather than removes this caveat [E003].

Plain perplexity-only distillation does not preserve brain alignment by default, and the headroom below the teacher is genuine but entangled with model quality. Whether an alignment-guided objective can recover that headroom beyond what perplexity already implies is the matched-perplexity question the rest of the program takes up; this screen does not settle that question. The E003 evaluation is a screen-level ROI benchmark (NC $\approx 0.35$) at matched compute budget. The screen is not powered to resolve a $\sim 0.005$ objective-specific gap and was not run at matched perplexity, so that question moves to the powered voxelwise substrate [E003][L011].

### 4.3 An alignment-guided objective has no demonstrated usable lever

**[GAP: Results Q2 (source E004), to write in Round 3: optimising $\mathcal{L}_{\text{brain}}$ on the strong aligner shows only a small apparent held-out alignment movement; the effect is entangled with perplexity, and at the honest fold-level inference unit the interval includes zero, so no usable lever is demonstrated (undemonstrated, not proven zero).]**

### 4.4 No per-individual gain at matched perplexity, and the averaging artifact

**[GAP: Results Q3 awaits finding-reports R09–R13 (sources E005, E008, E011, E013, E017, E019, E020): the per-individual null, the cross-subject-averaging artifact, the method-general failure, the quality law, the stimulus-predictability ceiling, and the external reproduction. The parked draft reports/_pending-Q3 reclaims R09.]**

### 4.5 No practical payoff

**[GAP: Results Q4/A3 awaits finding-report R14 (source E009): the bounded OOD-perplexity null and its null-by-construction reading.]**

Missing figure: ../figures/fig_lever.png

Figure 6: [PLACEHOLDER: generate from [E004]] The brain-specific lever on the strong aligner: held-out unique-$R^2$ change from optimising $\mathcal{L}_{\text{brain}}$ against its permuted twin control, shown at the honest fold-level inference unit (folds, not pooled cells), with the interval spanning zero and the single dominant fold marked. **[GAP: Round-3: render from the E004 record and finalise the caption.]**

## 5 Discussion

**[GAP: Discussion awaits the consolidated Results arc (R07–R14). The intended spine: the positive contribution is a measurement-validity finding plus an artifact-resistant protocol, not a bare null; cross-subject target averaging manufactures apparent brain-specificity, and the matched-perplexity, per-kind permuted-twin, per-subject protocol is the control the alignment-training literature lacks. Write only once the Results sections it interprets exist.]**

## 6 Limitations

**[GAP: Limitations awaits the consolidated Results arc. The intended content, each tied to its result: the absolute unique variance is a few percent of the noise ceiling (a scale fact, not a weakness); the reality claim is "beyond the nuisance variables actually subtracted," with surprisal held out by design; the per-individual null is measured at one power level on one cohort; the stimulus-predictability ceiling is instrument-limited rather than closed. Write once the results exist.]**

## 7 Conclusion

**[GAP: Conclusion awaits the consolidated Results arc. It should restate the established positive (the alignment signal is real and powered beyond low-level artifacts) and the program's net finding once R07–R14 are consolidated, without reaching past the report layer.]**

## Checkpoint log

### 2026-06-16: Checkpoint 1, the manuscript is opened

This is the first checkpoint: the extended manuscript is created and the layers that already trace to a written finding-report are drafted. The science behind it was produced over the preceding working sessions and recorded in the experiment files; this checkpoint consolidates the part of that record that the report layer has so far narrated.

What is folded in at this checkpoint. The measurement apparatus and the one established positive result: a trained language model's middle layer carries unique variance about real language-network activity beyond a low-level nuisance regression, and a same-architecture untrained network does not [E006]. This is consolidated from the finding-report on the reality of the signal (R06), which itself rests on the ROI screen [E002] and the powered voxelwise run [E006]. The front matter (Introduction, Related work) is consolidated from the background reports: the first-principles case for the brain as a weak prior (R03), the gap analysis that locates the contribution in the empty *shrink* × *optimize* cell (R04), and the synthesis of the measurement literature (R01).

What is deliberately left open. The Results sections beyond the reality of the signal (whether plain distillation preserves alignment, whether the loss is a usable lever, the per-individual null, the quality law, the predictability ceiling, and the practical-payoff question) are marked with `\gap` because their finding-reports (R07–R14) are not yet written. Knowledge flows down only: a number is narrated in a report before it is consolidated here, so these sections wait on the report layer rather than reaching past it into the raw experiment records. The abstract is provisional for the same reason: it can state the question and the reality result, but not yet summarise an arc whose later findings have no narration layer.

How this checkpoint was built. The draft was written through the scientific-writing skill on the full loop, with the review pass run continuously: adversarial critique agents argued against the reasoning and the prose quality section by section, and the objections that held were addressed before the section was kept.

### 2026-06-22: Checkpoint 2, the distillation-preservation result and a register repair

This checkpoint does two things, neither of which changes a recorded number: it folds in the next finding-report, and it repairs the prose register the first checkpoint shipped.

What is folded in. The Q1 result, that plain perplexity-only knowledge distillation does not preserve a teacher's brain alignment by default, is consolidated from its finding-report (R07, source [E003]) into the second Results subsection. A from-scratch logit-KD student lands a gap of +0.0181 below its teacher ($p < 0.001$, retaining 0.37 [0.14, 0.58] of the teacher's alignment above the untrained floor), and unique $R^2$ falls along a monotone gradient ordered by how far each representation was rebuilt away from conventional pretraining [E003]. The stronger claim, that the distillation objective sheds alignment beyond the language-model quality it costs, is left unsupported in the prose, because alignment co-varies with log-perplexity at Pearson −0.88 and the two objective-specific dissociations are only marginal [E003][L011]. The abstract and Introduction are updated to state this second result with that hedge, and their downstream-summary `\gap` markers are narrowed from R07–R14 to R08–R14. The glossary gains the terms the new section needs (knowledge distillation, cold-init/warm-init, headroom $\Delta$, floor-anchored retention $\rho'$, the data-processing inequality); the retention symbol $\rho'$ is disambiguated from the partial correlation $\rho_{XY \cdot Z}$ of the methods, a notation collision the consolidation guard flagged.

The register repair. A supervisor reading the v0.1 body flagged a storytelling register that the writing-style rules forbid (L054, D046): abstractions handed verbs of agency, the rhetorical move narrated instead of made, the repo's internal scaffolding metaphors leaking into readerfacing prose, and a reflex passive density above the threshold in the Introduction and Related work. This checkpoint rewrites those passages in the Introduction, abstract, Related work, and Results to assert the content directly, with a subject that acts and a verb that states, and varies the passive where the agent is the real topic. No claim, number, or hedge changed under the rewrite; only the voice did.

What is still open. The Discussion, Limitations, and Conclusion remain `\gap` because the results they interpret (the per-individual null, the averaging artifact, the quality law, the ceiling, the practical-payoff question) await their finding-reports R08–R14. R07's own caveats are carried inside the Results prose, as the report convention requires, rather than opened as a partial Limitations section ahead of the consolidated arc.

How this checkpoint was built. Written as its own logged `/write` session on the full loop (the ship gate D046 forbids folding a manuscript revision into another session), with the deterministic checks run on each edit and the non-skippable review pass, the `prose-register-auditor` included, run before handoff.

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